Anas M. Wagih

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Summary

Senior-year Electronics & Communication Engineering student (graduating 2027) with hands-on experience building AI-powered products and end-to-end ML pipelines. Proficient in Python, PyTorch, TensorFlow, and Generative AI APIs (LLMs/GPT), with practical data engineering experience using **NumPy**, **PIL**, and cloud-based training pipelines across open-source projects on GitHub. Passionate about applying ML and GenAI to solve real-world engineering problems at scale.

Education

Arab Academy for Science, Technology, and Maritime Transport — Egypt *09/2022 – 02/2027*
B.Sc. in Electronics and Communication Engineering

- **ABET**-accredited Degree, USA
- **Relevant Coursework:** Machine Learning, Deep Learning, Probability & Statistics, Linear Algebra, Digital Signal Processing, Microprocessors, Embedded Systems, Electrical Circuit Analysis

Experience

Software & IoT Engineering Intern — Clixsys *07/2025 – Ongoing*

- Built an **AI-powered in-app assistant** integrating **OpenAI's GPT API** (LLM) for natural-language smart-home control and real-time data retrieval; achieved ≈**2 s** end-to-end response latency with multi-voice support.
- Implemented a **Facial Recognition** pipeline (one-shot learning from user-profile images) for automated gate access, optimised for high accuracy and fast inference on edge hardware.
- Co-developed 3 cross-platform (iOS/Android) applications serving **10k+ active users** using Quasar (frontend) and Firebase/AWS (backend).

Embedded Systems Trainee — National Telecommunication Institute *08/2025 – 09/2025*

- Hands-on embedded C programming for AVR microcontrollers: GPIO, ADC, timers, interrupts, USART, SPI, I2C.
- Designed a Dual-MCU STM32-based Deadman Safety and Telemetry System for Electric Vehicles.

Projects

BitBlock — Open-Source No-Code Microcontroller Programming Platform [**Live Demo**]

- Engineered an end-to-end **on-device ML pipeline**: dataset collection → model training (Python / TensorFlow/Keras) → **INT8 post-training quantisation** → TFLite conversion and C header generation for microcontroller deployment, with hardware-gated model compatibility enforced per board RAM/flash constraints.
- Implemented sensor data preprocessing and temporal windowing using **NumPy** and **PIL** (image pipeline), optimised for efficient integration with the cloud-based TensorFlow training loop running on Google Cloud Run.
- Architected a browser-based **Blockly** editor that transpiles visual blocks into compiled C/C++ firmware for **10 ESP32/Arduino boards** — zero install, cloud-compiled via containerised toolchains on Google Cloud Run.
- Applied **software engineering best practices**: modular OOP design (React/TypeScript frontend, Dockerized backend), automated compilation testing, CI/CD pipeline — all open-source.

PCBs Portfolio — Open-Source Hardware (10+ Boards) [**Repo**]

- Designed and published production-ready **PCBs** — custom **ESP32/ESP8266** and **STM32** boards with optimised RF layout and power regulation — under *8bitlab*, a personal open-hardware initiative.

- Followed professional practices (DRC, footprint validation, layer stackup); released full schematics, Gerber files, and BOMs for direct fabrication.

Skills

ML / AI: PyTorch, TensorFlow, TensorFlow Lite, Keras, Scikit-learn, NumPy, PIL/Pillow, Pandas, Matplotlib, LLM / GenAI APIs (OpenAI GPT), NLP fundamentals, INT8 quantisation, on-device inference, RAG (conceptual).

Programming: Python, Embedded C, JavaScript, MATLAB, C++, SQL.

Tools & Cloud: Git, AWS, Firebase, Google Cloud Run, Linux, Microchip Studio, STM32CubeIDE, KiCad, Android Studio, Xcode, Arm Mbed.

Hardware: Arduino, Raspberry Pi, ESP32, PCB Design, Power Electronics, Soldering.

Languages: Arabic (Native) & English (Proficient — Cambridge IGCSE O-Level: **"A" grade**)

Certifications

- **DeepLearning.AI + Stanford University — ML Specialization (3 Courses)** [**Link**]
- **DeepLearning.AI — Deep Learning Specialization (5 Courses)**
- **Arm Education — Arm Cortex-M Architecture & Software Development (4 Courses)** [**Link**]
- **Arm Education — Advanced Embedded Systems on Arm (2 Courses)** [**Link**]

Activities

Machine Learning Tutor — ECE Department ML Workshop *(Linear & Logistic Regression, KNN)*

Arduino Co-Tutor — ECE Department hands-on Arduino Workshop

University Robotics Club — Electrical Team Leader